BAC Stress Test: Full Methodology and Documentation

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The project projects Bank of America's credit losses, pre-provision net revenue (PPNR), loan-loss allowance and common equity tier 1 (CET1) ratio over nine quarters (2026Q1-2028Q1) under the Federal Reserve's 2026 Baseline and Severely Adverse scenarios, using public data only. In the severe case, applying the Fed's published allowance and capital-action rules, it ends at 10.30% CET1 (minimum 10.26%) against the Fed's published 10.8% (minimum 9.9%), with PPNR of $67.6B against the Fed's $66.2B; the rest of this document says where each number comes from and why.

1. Purpose, scope and how to read this document

The Fed publishes its own result for Bank of America in the DFAST 2026 results (June 24, 2026). That result is used only as an outside benchmark. Nothing in the model is fitted to it.

The pipeline has six blocks, each with its own section below:

1. **Credit losses** (sections 4 and 5): industry charge-off rates by loan segment, projected under each scenario, then scaled to Bank of America's own loan mix and loss history.
2. **PPNR** (section 6): net interest income, fee income and expense equations on Bank of America's income statement, plus stress overlays.
3. **NII rate sensitivity** (section 7): a balance-sheet model fitted to the sensitivities Bank of America discloses in its 10-Ks; it supplies the interest-rate effect on PPNR.
4. **Basel EVE** (section 8): economic value of equity under the six Basel standardised shocks. Bank of America does not disclose EVE, so this block rests on assumptions and stays illustrative.
5. **Allowance and CET1 waterfall** (section 9): a CECL-style reserve and the capital bridge from the 2025Q4 starting point.
6. **Validation and results** (sections 10 to 13): comparison with the Fed, assumptions register, limitations.

Section 14 lists how to refine the work and section 15 maps files to run order. Conventions: dollar amounts are in billions unless stated; rates are annualised percentages; quarters are calendar quarters (2026Q1 = January to March 2026); "severe" means the Fed's

Severely Adverse scenario.

The work is a self-built illustration. It has not been independently validated, is not a regulatory submission, and should not be used for capital decisions.

2. Data sources and why each item is needed

Eight sources feed the model. The first table gives each source, what was taken from it and the reason; the following tables itemise the Fed scenario variables, the SEC tags and the FR Y-9C items.

Source	What was taken	Why it is needed
Fed 2026 supervisory scenario files (Historic, Baseline, Severely Adverse; domestic)	16 macro and market variables, quarterly: history 1976Q1-2025Q4, scenario paths 2026Q1-2029Q1 (13 quarters)	Drivers of the loss models (unemployment, spreads, house and CRE prices), the equity path for the fee stress, and the rate path for the NII effect. History and projection share definitions, so the models are estimated and projected on the same variables
FRED, Federal Reserve charge-off and delinquency rates, top-100 banks	Net charge-off rates for cards, business (C&I), CRE excluding farmland, single-family mortgage and other consumer loans; delinquency rates for cards, business, CRE and mortgage; quarterly from 2000Q1	Industry loss history long enough to include the 2008 crisis and COVID, which Bank of America's own history cannot supply. Script prefers the top-100-bank series and falls back to all-commercial-bank series; the audit file fred_audit.csv, written when the fetch script runs, records which resolved

Source	What was taken	Why it is needed
SEC XBRL company facts, Bank of America (CIK 70858)	Quarterly net interest income, noninterest income and expense, net income, loan and unfunded provisions, charge-offs, recoveries, allowance, total assets, loans	Bank of America's own income statement for the PPNR equations and its own charge-offs to scale the industry curves
FFIEC FR Y-9C, Bank of America (RSSD 1073757)	Year-end 2025 (loan mix, CET1, risk-weighted assets, allowance, dividends, tax) and 2023-2025 Q3 and Q4 balance-sheet detail for the rate-risk models	The capital starting point must match the Fed's starting point; repricing and maturity detail is not in the SEC data
Bank of America Form 10-K, FY2023, FY2024, FY2025	Net interest income sensitivity tables (Table 44 static deposits, Table 45 dynamic deposits), long-term debt note, fair value of financial instruments, fair-value-hedge table	The only disclosed interest-rate sensitivities, used to fit the NII model, and the data on hedges that convert fixed items to floating
Bank of America 10-K segment tables 2007-2009 and 2019-2025, plus 4Q22 and 4Q25 earnings presentations	Investment banking fees (total corporation), Global Wealth and Investment Management fee lines and assets under management, sales and trading revenue, fee mix of noninterest income	Sourcing the fee stress overlays (equity pass-through, investment banking haircut, sales and trading haircut)
Fed DFAST 2026 results (June 24, 2026) and data dictionaries	Bank of America severe-scenario result, Table 10 loss rates, aggregate operational-risk losses and PPNR	Benchmark only, plus the operational-risk overlay and the Fed-set securities, trading and other losses included in the waterfall

Source	What was taken	Why it is needed
Basel Committee BCBS d368 (April 2016)	Standardised EVE shocks, time buckets, non-maturity-deposit and prepayment parameters	The only specification for the EVE calculation, since Bank of America does not disclose EVE

Fed scenario variables (16). Real GDP growth, nominal GDP growth, real disposable income growth, nominal disposable income growth, unemployment rate, CPI inflation, 3-month Treasury rate, 5-year Treasury yield, 10-year Treasury yield, BBB corporate yield, mortgage rate, prime rate, Dow Jones Total Stock Market Index, house price index, commercial real estate price index and the VIX. The BBB spread is the BBB yield less the 10-year yield. The loss models use unemployment, the BBB spread, house prices and CRE prices; the fee overlay uses the equity index; the NII model uses the 3-month, 5-year and 10-year rates.

SEC XBRL tags (quarterly, $ billions). Where Bank of America changed tags, the script stitches a chain of tags, newest first, so a later tag fills only quarters the earlier ones lack.

Concept	Tags used	Treatment
Net interest income	InterestIncomeExpenseNet	3-month facts used directly; missing quarters derived from year-to-date facts (6M less 3M, 9M less 6M, 12M less 9M)
Noninterest income, expense, net income	NoninterestIncome, NoninterestExpense, NetIncomeLoss	Same

Concept	Tags used	Treatment
Loan provision	FinancingReceivableExcludingAccruedInterestCreditLossExpenseReversal, then ProvisionForLoanLossesExpensed, then ProvisionForLoanLeaseAndOtherLosses	Loans-only from 2019; the older tag also includes unfunded commitments, so unfunded provision is carried separately
Charge-offs, recoveries	AllowanceForCreditLossWriteoff and Recovery tags (three generations each)	Old recoveries tag is negative from 2011, so its absolute value is used
Allowance	FinancingReceivableAllowanceForCreditLossExcludingAccruedInterest and older tags	One-time opening balances dated 12/31/2019 (CECL adoption) and 12/31/2009 (consolidation of off-balance-sheet vehicles) are dropped
Total assets	Assets	Used for the operational-risk overlay and the PPNR equations

CET1 capital and risk-weighted assets are not tagged consistently for Bank of America in the SEC data, so they come from the Y-9C.

FR Y-9C items (values in $ thousands in the file; converted to $ billions).

Items	Description	Used for
BHCAP793, BHCAA223	CET1 ratio (11.3602%); total risk-weighted assets ($1,772.9B)	Starting CET1 = ratio x RWA = $201.4B
BHCA8274	Tier 1 capital ($227.4B)	EVE outlier test (15% of Tier 1)

Items	Description	Used for
BHCKB528, BHCK2122	Loans held for investment ($1,243.5B); total loans held for investment and sale ($1,248.7B)	Allowance base (B528); loan-mix denominator (2122)
BHCK3123	Allowance for credit losses on loans and leases ($13.2B)	Opening allowance; calibrates the CECL horizon
BHCK4460, BHCK4598	Common dividends declared ($8.083B); preferred dividends ($1.445B)	Capital distributions in the waterfall
BHCK4301, BHCK4302	Pre-tax income ($37.7B); income taxes ($7.2B)	Effective tax rate 19.1%
BHCKB538	Credit card loans	Loan mix: cards
BHCK1763, BHCK1764	C&I loans to U.S. and non-U.S. addressees	Loan mix: business
BHCKF158-F161	Construction, land, and owner-occupied and other nonfarm nonresidential real estate loans	Loan mix: CRE
BHCK1410 less CRE	Loans secured by real estate, residual after CRE (includes 1-4 family, HELOC, multifamily)	Loan mix: mortgage
BHCKK137, K207, B539	Automobile, all other consumer and other revolving credit loans	Loan mix: other consumer
BHDM5367, BHDM5368, BHDM1797	First-lien, junior-lien and revolving (HELOC) loans secured by 1-4 family	Floating versus fixed loan split in the rate models
BHDM6631, BHDM6636	Noninterest-bearing deposits ($517.8B); interest-bearing deposits ($1,361.2B)	Deposit base for NII and EVE
BHCK1771, 1772, 1773, 1754	Held-to-maturity fair value; available-for-sale amortised cost and fair value; amortised cost total	Securities repricing and EVE valuation

Items	Description	Used for
BHCK0383, 0384, 0387	Debt securities by remaining maturity (up to 1, 1-5, over 5 years)	Maturity ladder for securities
BHCK2332, 2333, 4062	Other borrowed money up to 1 year ($154.8B) and over 1 year ($250.4B); subordinated notes ($27.6B)	Wholesale funding repricing
BHCKB989, B995	Reverse repos ($316.6B) and repos ($344.7B)	Short-term asset and liability repricing

The Y-9C also carries fiduciary income (BHCK4070, $2.6B), securities brokerage fees (C886, $17.1B), investment banking fees (C888, $5.3B) and trading revenue (A220, $12.8B) for 2025. These were reviewed as a cross-check but are not model inputs, because three calm years cannot identify a stress response.

3. Time frames and data constraints

Each block uses the longest window its own data supports; the strict common window across every series is only 2011Q1-2025Q4 (60 quarters), limited by Bank of America's net charge-offs, so a single common window would have thrown away the 2008 and 2000-2010 loss history.

Block	Window	Observations	Why this window
Fed scenario history and paths	History 1976Q1-2025Q4; projection 2026Q1-2029Q1	200 + 13 quarters	Same variable definitions in history and projection; only nine projection quarters (2026Q1-2028Q1) are used for results because the Fed's published result covers nine

Block	Window	Observations	Why this window
Segment loss models	2000Q1-2025Q4	104 for cards, business, other consumer; 66 CRE; 71 mortgage	Includes the 2008 crisis and COVID. CRE and mortgage drop quarters with net charge-off at or below 3 basis points (net recoveries), where the log is undefined
COVID adjustment	Dummy 2020Q2-2021Q4	7 quarters	Government support suppressed losses; without the dummy it distorts the persistence estimate
Bank of America charge-offs	2011Q1-2025Q4	60 quarters	Start set by the first quarter with consistent XBRL charge-off tags
BAC scale factor fit	2014Q1-2019Q4	24 quarters	2011-2013 excluded: Bank of America's legacy Countrywide-era book ran at 1.69 times the industry-implied rate
BAC scale factor test	2020Q1-2025Q4	24 quarters	Out-of-sample check
PPNR equations	2014Q1-2025Q4 (equations refit to 2019 and tested on 2020-2025)	48 quarters	2008-2013 is dominated by one-offs (goodwill impairment, litigation, run-off). Income data exist from 2008Q2 (73 quarters)
NII fit	10-K disclosures for year-end 2023, 2024 and 2025	30 disclosed points	Only three years of sensitivity tables were provided and used; out-of-sample test refits without 2025

Block	Window	Observations	Why this window
Fee overlay evidence	2007-2009, 2019-2025, plus quarterly 4Q21-4Q22	Annual segment tables	Covers the 2008 crisis, 2020 and 2022; no quarterly series beyond 4Q22
EVE	Balance sheet at 2025Q4 (Q3 for flows)	One date	Basel EVE is a point-in-time measure
Allowance and CET1	Start 2025Q4; projection 2026Q1-2028Q1	9 quarters	Matches the Fed's starting point and horizon

Data constraints that shape the results.

- **Short Bank of America history.** Bank of America's own losses have not been stressed since the 2008-2011 period, which carries the Countrywide legacy and cannot be used. The industry curves supply the stress shape; Bank of America supplies one scale factor.
- **Industry series are not bank-specific.** FRED reports industry charge-off rates for the top 100 banks; Bank of America's segment mix differs, and its CRE and mortgage books behave differently from the industry average (see the segment misses in section 10).
- **Mortgage grouping.** The Y-9C extract does not split residential from multifamily, so the mortgage bucket is first-lien, junior-lien, HELOC and multifamily together; the Fed reports first-lien mortgages separately, so the comparison is approximate.
- **Loan base differences.** Total loans are $1,248.7B (Y-9C held for investment and sale), $1,243.5B (held for investment, used for the allowance and loss dollars) and $1,149.1B (10-K carrying amount in the fair-value footnote). These are definition differences and were left unreconciled; losses in dollars use the held-for-investment figure.
- **SEC data quality.** XBRL tags changed over time and some periods were restated across filings; the stitching script reports overlaps. The sandbox used for development could not reach SEC, FRED, Fed or BIS sites, so the fetch scripts are written to run on a local machine and the data files were supplied by the user.
- **No Adverse scenario from the Fed.** The Fed publishes Baseline and Severely Adverse only. An Adverse case is constructed halfway between them (price indices interpolated in logs) and labelled as an assumption.

- **Y-9C coverage.** Six Y-9C files were available (2023Q3, 2023Q4, 2024Q3, 2024Q4, 2025Q3, 2025Q4), enough for the 2025 starting point and the rate-risk fits but not for a long history of loan-type charge-offs, which would need about 60 quarterly files.
- **Disclosure gaps.** Bank of America does not disclose economic value of equity, its deposit behavioural assumptions or its prepayment speeds, so these are assumptions.

4. Credit loss models

Each of five loan segments (cards, business, CRE, mortgage, other consumer) has its own equation for the log of the industry net charge-off rate, estimated on 2000Q1-2025Q4 and projected under each Fed scenario. The model is the unrestricted one-step error-correction form, which is algebraically an ARDL(1,1): losses depend on last quarter's losses plus the current and lagged drivers.

Why this form. A first version used a two-step Engle-Granger error-correction model on delinquency against unemployment levels. The data rejected it: delinquency is close to a unit root (persistence 0.94 to 1.01, residual AR(1) 0.85 to 0.97) and the error-correction terms had the wrong sign. The one-step form avoids imposing a cointegrating relationship and gives a sensible speed of adjustment for charge-offs.

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y_t = a + \rho\, y_{t-1} + b_0' x_t + b_1' x_{t-1} + c\, \mathrm{COVID}_t + e_t
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\Delta y_t = a + (\rho - 1)\, y_{t-1} + b_0' \Delta x_t + (b_0 + b_1)' x_{t-1} + c\, \mathrm{COVID}_t + e_t
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Here y is the log of the annualised net charge-off rate (LOSS equation) or the logit of the delinquency rate (PD equation); x is the driver set below; the speed of adjustment is 1 minus rho; the half-life of a shock is ln(0.5) / ln(rho) quarters.

Drivers. Derived from the Fed's macro variables:

Driver	Definition	Why
ur	Unemployment rate	Level effect on mortgage and CRE
urmin	Unemployment minus its trailing 12-quarter minimum (unemployment gap)	Captures the turn in the cycle, not the level; consumer and business losses respond to a rising gap

Driver	Definition	Why
dur4	Four-quarter change in unemployment	Speed of deterioration; cards and other consumer
spread	BBB yield minus 10-year Treasury yield	Credit conditions for business loans
hpi4	Four-quarter log change in the house price index	Collateral value for mortgages
cre4	Four-quarter log change in the CRE price index	Collateral value for CRE

Segment	Drivers (current and one lag)
Cards	urmin, dur4
Business	urmin, spread
Other consumer	urmin, dur4
CRE	ur, cre4
Mortgage	ur, hpi4

Estimation. Ordinary least squares with Newey-West standard errors (4 lags), a COVID dummy for 2020Q2-2021Q4 (not identified for mortgage, whose COVID quarters fall below the 3 basis point floor and are excluded), and a floor of 3 basis points (0.0003): CRE and mortgage quarters with net charge-offs at or below the floor are excluded from the log fit and the projection is floored there.

Fit and diagnostics (LOSS equations).

Segment	Observations	R2	Persistence rho	Half-life (quarters)	Residual AR(1)	Residual s.d. (log)
Cards	104	0.935	0.740	2.3	-0.169	0.100
Business (C&I)	104	0.941	0.890	6.0	-0.018	0.202
Other consumer	104	0.900	0.836	3.9	-0.130	0.158
CRE	66	0.900	0.865	4.8	-0.001	0.422
Mortgage	71	0.938	0.780	2.8	-0.116	0.328

Residuals are not autocorrelated and shocks decay within 2 to 6 quarters, so the dynamics are well behaved. Weak spots: the CRE equation's unemployment terms have the wrong sign and are insignificant (current -0.12, lag +0.11), and the CRE price term (-1.87, t -2.1) carries the whole stress response; the mortgage house-price term (-2.82, t -1.7) is borderline. These are why the CRE and mortgage curves miss the Fed's (section 10).

Delinquency (PD proxy) equations were estimated for cards, business, CRE and mortgage and then rejected as a loss engine: persistence of 0.94 to 1.01, speeds of adjustment near zero (0.02 for cards, 0.03 for business, negative for CRE) and half-lives of 10 to 36 quarters mean they behave like random walks. They remain in the output files as a diagnostic, and an implied LGD (net charge-offs this quarter divided by last quarter's delinquency) is reported but not forecast.

Projection. For each scenario the model is run recursively from the last actual rate (2025Q4): the first projected log rate uses the 2025Q4 actual as its lag, each later quarter uses the previous projection, the scenario's drivers and the COVID dummy set to zero. The scenario drivers are winsorised at the minimum and maximum seen in the 2000Q1-2025Q4 fit window, so no driver is extrapolated beyond history; a no-cap variant is saved for comparison. The path is the exponential of the fitted log rate with no variance correction, so it is a median path, slightly below the mean.

Result of the cap. The caps do not bind in any scenario, so the severe case is not driven by extrapolating a single variable beyond history.

5. Scaling the industry curves to Bank of America

The industry curves are scaled to Bank of America with one factor, k = 0.955, fitted on Bank of America's own 2014Q1-2019Q4 charge-offs and then tested on 2020Q1-2025Q4, where it explains 72% of the variance with an error of 0.07 percentage points. One factor is used because Bank of America's own history has too little stress to estimate segment-level scales.

Step 1: loan mix. Segment weights come from the 2025Q4 Y-9C as a share of total loans ($1,248.7B). The five segments cover 73% of loans; the other 27% (loans to non-bank financials, foreign governments, other) gets no industry curve, so the scale factor absorbs them.

Segment	Weight	Y-9C items
Cards	8.5%	BHCKB538
Business (C&I)	29.0%	BHCK1763 + BHCK1764
CRE	5.6%	BHCKF158 to F161
Mortgage (first lien, junior lien, HELOC, multifamily)	22.3%	BHCK1410 less the CRE items
Other consumer	7.2%	BHCKK137 + K207 + B539

Step 2: implied rate. Industry-implied Bank of America charge-off rate each quarter, from FRED:

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\hat r_t = \sum_i w_i \, r_{i,t}
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Step 3: actual rate. From the SEC data, net charge-offs annualised over average loans:

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r^{BAC}_t = \frac{4 \times NCO_t}{(L_t + L_{t-1})/2} \times 100
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Step 4: scale factor. Least squares through the origin on the fit window:

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k = \frac{\sum_t r^{BAC}_t \, \hat r_t}{\sum_t \hat r_t^{\,2}}
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Step 5: projection. The projected Bank of America rate is k times the weighted sum of the projected segment rates, applied to loans held flat at the latest level; dollars are the quarterly rate (annual rate divided by 4) times held-for-investment loans of $1,243.5B. The Fed uses its own balance path, so this is a simplification.

Why this window. 2011-2013 is excluded because actual losses ran at 1.69 times the industry-implied level: the legacy Countrywide book was still running off, and the current mix is not what produced those losses. Fitting only on 2014-2019 uses a benign period, so the in-sample R2 (-0.38) is not meaningful: Bank of America's rate barely moved, so there is no variance to explain. The out-of-sample test on 2020-2025 is the real check.

Check	Result
k, fit 2014Q1-2019Q4	0.955
k, 2014Q1-2025Q4	0.896

Check	Result
Mean implied vs actual rate, 2014-2025	0.46% vs 0.42%
Out-of-sample R2, 2020-2025	0.72
Out-of-sample RMSE and bias	0.07 and -0.05 percentage points

Sensitivity of k to the mix assumption. The fitted factor moves with the weights:

Mix assumption	k
Base	0.955
Cards weight +20%	0.849
Cards weight -20%	1.091
Business weight +20%	0.919
CRE weight +50%	0.955
Mortgage weight -20%	0.964

Cards matter most because they have the highest loss rate. CRE barely moves k in the fit window because CRE losses were near zero in 2014-2019, which is also why the model cannot be tested on CRE stress.

Outcome. Under the severe scenario Bank of America's nine-quarter cumulative loss is 4.7% of loans (about $59.3B), against the Fed's 5.2% ($64.9B). Only this total is scaled to Bank of America; segment rates in section 10 are industry curves and are shown beside the Fed's, not tuned.

6. PPNR model and stress overlays

PPNR is net interest income plus noninterest income minus noninterest expense, projected from three equations estimated on Bank of America's own income statement (2014Q1-2025Q4) and then adjusted by stress overlays that are applied to the severe and constructed Adverse cases only. The data-driven model alone gives $112.6B of nine-quarter severe PPNR, far above the Fed's $66.2B, because the 2014-2025 sample contains no deep recession; the overlays bring it to $67.6B.

Equations. Each is the same ARDL(1,1) form as the loss models, with a COVID dummy for 2020Q2-2021Q4 and total assets held flat at $3,412B:

Equation	Dependent variable	Drivers (current and lag)	Why these drivers
NIM	NII x 4 / total assets, %	3-month Treasury rate; slope (10-year less 3-month)	Margin responds to the policy rate and the curve
FEE	Noninterest income x 4 / total assets, %	4-quarter log change in equity index; log VIX	Wealth and trading fees follow markets
EXP	Log noninterest expense	Log revenue (NII plus noninterest income)	Expenses flex with revenue; long-run elasticity 0.64

```
y_t = a + \rho\, y_{t-1} + b_0' x_t + b_1' x_{t-1} + c\, \mathrm{COVID}_t + e_t
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NII_t = \frac{NIM_t}{100} \cdot \frac{A}{4}, \qquad NONII_t = \frac{FEE_t}{100} \cdot \frac{A}{4}, \qquad PPNR_t = NII_t + NONII_t - \exp(\ln EXP_t)
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The long-run expense elasticity to revenue is (b_lrev + b_lrev_l1) / (1 - rho) = (0.5889 - 0.4096) / (1 - 0.721) = 0.64.

Fit and out-of-sample test. Estimated 2014Q1-2025Q4 (48 quarters); the out-of-sample check refits before 2020Q1 and tests on 2020-2025:

Equation	In-sample R2	Persistence rho	Out-of-sample R2	Out-of-sample RMSE
NIM	0.876	0.784	-14.2	0.66 pp
FEE	0.618	0.749	-7.6	0.42 pp
EXP	0.668	0.721	0.10	0.08 (log)

The margin and fee equations fit in sample but fail out of sample: the 2014-2019 training sample never saw zero rates or the 2020-2021 deposit surge. This is why the stress response is taken from sourced overlays and the NII model, not from the equations.

Why 2014Q1 start. Income data exist from 2008Q2, but 2008-2013 is dominated by items that are not macro-driven: the Countrywide and Merrill integrations, a $10B goodwill impairment in 2010 and mortgage representation, warranty and litigation charges in 2010-2013. The NIM equation fitted from 2008Q4 has R2 0.47 against 0.87 from 2014Q1, and the fee and expense lines contain one-offs of $10-20B in single quarters.

Stress overlays. Applied on top of the Baseline-path equations, quarter by quarter, over the severe path:

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NII^{s}_t = NII^{b}_t + \Delta NII_t
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FeeLoss_t = NONII^{b}_t \left[ w_E\, \varepsilon\, g_t + \left( w_{IB}\,
h_{IB} + w_{ST}\, h_{ST} \right) \frac{g_t}{g_{max}} \right]
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g_t = \max\!\left(0,\; 1 - \frac{Equity^{severe}_t}
{Equity^{baseline}_t}\right), \qquad EXP^{s}_t = EXP^{b}_t \left(
\frac{Rev^{s}_t}{Rev^{b}_t} \right)^{0.64} + OpRisk_t
```

Here Delta NII comes from the rate-risk model (section 7); g is the equity gap versus Baseline and g_max its severe peak, so the investment banking and sales and trading haircuts apply in full at the worst quarter and scale down with the gap; the other terms are in the table below. Revenue falls, so expenses fall with the 0.64 elasticity, and operational-risk losses are added to expense.

Parameter	Value	Source and basis
NII effect (Delta NII)	Quarterly series, dynamic deposits	build_nii_eve_model.py: model fitted to Bank of America 10-K Tables 44 and 45; effect uses the gap between the severe and Baseline 3-month rate each quarter
w_E, equity-linked fee share	33.2% (= 17.6 / 53.0)	4Q25 earnings presentation: asset management $15.3B plus brokerage $2.3B of $53.0B FY2025 noninterest income
w_IB, investment banking fee share	12.5% (= 6.6 / 53.0)	Same deck: investment banking fees $6.6B

Parameter	Value	Source and basis
w_ST, sales and trading share	39.4% (= 20.9 / 53.0)	Same deck: sales and trading $20.9B
epsilon, equity pass-through	0.6 (range 0.5 to 0.7)	10-K GWIM 2007-2009: Columbia Management fees $1,435M (2007) to $1,090M (2009), -24% against -34% for the Fed's equity index (0.7); U.S. Trust 2008 to 2009 -10% against -16% (0.6); GWIM assets under management -19% in 2008 against a 39% equity fall (0.5)
h_IB, investment banking haircut at peak	40% (range 24% to 54%)	Total corporation investment banking fees 4Q22 $1,071M against $2,351M a year earlier (-54%, boom base) or against the 2019 run-rate of $5,642M / 4 (-24%); fees rose in 2008 (+7%, $2,537M to $2,708M) and 2020 (+27%, $5,642M to $7,180M), so only 2022 supports a fall
h_ST, sales and trading haircut at peak	0%	Client revenue did not fall in stress (4Q22 $3,525M against $2,936M a year earlier; 2009 up). The 2007-2008 losses (-$2.1B, -$5.8B) were mark-to-market on legacy structured products, which the Fed treats as the separate trading and counterparty loss, not PPNR
OpRisk, operational-risk loss	0.88% of assets over 9 quarters ($30.0B in total, about $3.3B a quarter)	Fed 2026 DFAST: aggregate operational-risk losses $211B; aggregate PPNR $719B equals 3% of average assets, so $211B / ($719B / 0.03) = 0.88%; spread evenly over the first nine quarters and added to expense

Treatment of Fed-set trading losses. Trading and counterparty losses are not part of the Fed's PPNR; they enter the capital waterfall separately (section 9).

Constructed Adverse. The same overlays are applied with the gaps scaled by the Adverse scenario's own (smaller) equity and rate gaps, relative to the severe peak.

Correction history. An earlier version applied the NII rate sensitivity with the wrong sign (lower rates raised NII), which inflated severe PPNR to $82.1B and made pre-tax income appear to match the Fed's. Replacing it with the fitted NII model's effect fixed this; the offset of two errors had hidden the provisions gap.

7. NII rate-sensitivity model

The NII model reproduces Bank of America's disclosed 12-month NII sensitivities with an in-sample error of $0.2-0.3B and a held-out error of $0.4B, and supplies the interest-rate effect on stressed PPNR. It is a static-balance model of the 2025Q4 banking book, not Bank of America's internal model.

Why it was needed. The PPNR equations cannot be trusted for rate stress (their out-of-sample R2 is negative), and an earlier overlay with a hand-set sensitivity had the wrong sign. Bank of America discloses how its NII changes for parallel, flattening and steepening shifts, so a model built to match those 30 disclosed points ties the stress to the bank's own numbers.

Step 1: balances and rates. Average of Q3 and Q4 2025 Y-9C balances, with 4Q25 interest isolated as the Q4 year-to-date flow less the Q3 year-to-date flow. The rebuilt NII ties to the reported figure ($15.70B for 4Q25).

Category	Side	Average balance ($B)	4Q25 interest ($B)	Annualised rate (%)
Cash and interest-bearing balances	Asset	219.3	2.22	4.06
Reverse repo	Asset	321.2	3.76	4.69
Trading assets	Asset	407.9	3.14	3.08
Debt securities (AFS and HTM)	Asset	915.6	6.61	2.89
Loans and leases	Asset	1,241.3	17.70	5.71
Interest-bearing deposits	Liability	1,482.0	8.27	2.23
Repo	Liability	343.7	4.20	4.89

Category	Side	Average balance ($B)	4Q25 interest ($B)	Annualised rate (%)
Other borrowed money and trading liabilities	Liability	576.4	5.71	3.96
Subordinated debt	Liability	27.7	0.33	4.83

Step 2: repricing rules. A shock has a long-end part s (applied to the long curve) and a short-end part s_h (applied to funding and floating items); a parallel shock sets both equal. The 12-month NII change is assets less liabilities:

```
\Delta NII = \underbrace{S\, s_h + L\, \phi\, s_h + L(1-\phi)\,\tfrac{1}{T}\,\tfrac12\, s\, \lambda + R\,\tfrac12\, s\, \lambda}_{\Delta \text{assets}} \;-\; \underbrace{\Delta C_{dep} + Rp\, s_h + B_{ST}\, s_h + B_{LT}\, \theta\, s_h}_{\Delta \text{ liabilities}}
```

- S: cash, reverse repo and trading assets (reprice fully within the year).
- L: loans; phi = 66.3%: share assumed floating (C&I, cards, other revolving, loans to non-bank financials, other loans, HELOC and CRE; mortgages and auto treated as fixed). Balances are Y-9C data; the floating mapping is an assumption.
- T = 5 years: assumed average life of fixed loans, so 1/T of them reprice in a year; the 1/2 reflects reinvestment spread evenly over the 12 months, so the average new rate applies for half the year.
- R: securities repricing in the year = (up to 1 year bucket) x 1 + (1-5 year bucket) x min(1, 1/3) + (over 5 year bucket) x (1/9 + 10% prepayment). Buckets are Y-9C data ($101B, $302B, $506B); midpoints 0.5, 3 and 9 years and the 10% MBS prepayment are assumptions.
- lambda = 0.16: the long-end repricing scale, calibrated; it absorbs the pay-fixed hedges that make Bank of America's fixed-rate assets reprice far less than their contractual terms suggest.
- Rp: repo; B_ST: other borrowings up to one year plus trading liabilities (both reprice fully); B_LT: borrowings over one year, with theta = 100% effectively floating, calibrated.

Deposits. Interest-bearing deposits are split into a low-rate tier (share a, cost c_a = 0.15%, floored at zero) and the rest (cost c_b backed out so that the blended cost equals the observed 2.23%):

$$c_b = \max\!\left(\frac{r_{ib} - a\, c_a}{1 - a},\, 0\right), \qquad \Delta C_{dep} = D\left[a\,(\max(c_a + \beta s_h, 0) - c_a) + (1-a)\,(\max(c_b + \beta s_h, 0) - c_b) \right]$$

where beta is the deposit beta (share of the rate move passed to depositors) and the floor at zero makes down-shocks hurt more than up-shocks help.

Dynamic deposits. Bank of America discloses sensitivities with deposit balances held constant (static) and with balances allowed to migrate (dynamic). The gap is modelled as non-interest-bearing balances (NIB, $532B) leaving when rates rise and returning when they fall:

$$\text{Mig} = \begin{cases} \kappa_u \cdot NIB \cdot x^{\gamma_u} & x \ge 0 \\ \kappa_d \cdot NIB \cdot |x|^{\gamma_d} & x < 0 \end{cases}, \qquad x = s_h / 100\text{bp}$$

The fitted values are kappa_u = -0.00375 with gamma_u = 1.20 (cost of $2.0B per +100bp) and kappa_d = 0.0018 with gamma_d = 0.34 (concave benefit when rates fall). Dynamic NII change = static change + Mig.

Step 3: fitting. Bounded least squares (scipy) over all 30 disclosed points (2023 static: 6; 2024 static: 8; 2024 dynamic: 8; 2025 dynamic: 8) from the 10-K Tables 44 and 45. Deposit beta, low-rate tier share, long-debt floating share and the four migration parameters are shared across years; the long-end scale is shared too. Everything not fitted is data or an assumption flagged in nii_eve_assumptions.csv.

Parameter	Value	Status
Deposit beta	0.41	Calibrated
Low-rate deposit tier share	4.3%	Calibrated
Long-end repricing scale (net of hedges)	0.16	Calibrated
Floating share of borrowings over one year	100% (at the upper bound)	Calibrated; contractual floating is about 12% before swaps, so the fit is using the fair-value hedge swaps
Deposit migration	-$2.0B per +100bp	Calibrated
Floating share of loans	66.3%	Assumed mapping, data balances

Parameter	Value	Status
Fixed-loan life	5 years	Assumed
Securities prepayment on over-five-year bucket	10% a year	Assumed
Low-rate tier cost	0.15%	Assumed

Validation.

Test	Result
In-sample RMSE, static-deposit points (14)	$0.28B
In-sample RMSE, dynamic-deposit points (16)	$0.22B
Out-of-sample: fitted without 2025, predicting the 8 disclosed 2025 points	RMSE $0.41B
Direction check	NII falls when rates fall, rises when rates rise; the effect used for stress is the quarterly gap in the 3-month rate between severe and Baseline

The 2025 points, fitted and held-out:

2025 dynamic shock	Disclosed ($B)	Fitted ($B)	Held-out prediction ($B)
+100 parallel	0.7	0.93	1.13
-100 parallel	-2.0	-2.12	-2.25
+200 parallel	0.8	1.26	1.74
-200 parallel	-4.9	-5.05	-5.20
Flattener: short +100	0.5	0.62	0.79
Flattener: long -100	-0.3	-0.31	-0.34
Steepener: short -100	-1.7	-1.81	-1.91
Steepener: long +100	0.3	0.31	0.34

The weakest points are the up-200 basis point result (model overstates it, $1.26B fitted against $0.8B disclosed) and the 2023 down-100 results (model $0.6B too negative). Results for a parallel -300 basis point shock are extrapolated beyond anything disclosed.

Robustness. Refitting all four datasets after changing an assumed input moves the -100 basis point dynamic result between -2.07 and -2.18 and the +100 result between 0.89 and 0.99:

Assumption changed	Fit RMSE ($B)	Dynamic -100bp ($B)	Dynamic +100bp ($B)
Base fit	0.25	-2.12	0.93
Floating loan share -10pp	0.23	-2.07	0.89
Floating loan share +10pp	0.26	-2.13	0.94
Fixed-loan life 3 years	0.25	-2.12	0.93
Fixed-loan life 8 years	0.25	-2.12	0.93
Low-rate tier cost 0%	0.25	-2.12	0.94
MBS prepayment 5%	0.25	-2.12	0.94
Long debt 12% floating (contractual only)	0.33	-2.18	0.99

The fit prefers fully floating long debt (RMSE 0.25 against 0.33 with the contractual 12%), which is consistent with the fair-value-hedge swaps in the 10-K converting about $175.7B of fixed debt to floating.

Results (12 months, 2025 balance sheet, instantaneous parallel shift).

Shock	Static deposits ($B)	Dynamic deposits ($B)	Dynamic, % of run-rate NII
+200bp	5.86	1.26	2.0%
+100bp	2.93	0.93	1.5%
-100bp	-3.10	-2.12	-3.4%
-200bp	-6.29	-5.05	-8.0%
-300bp (extrapolated)	-9.48	-8.05	-12.8%

Scenario effect used in PPNR. For each quarter the model takes the gap between the scenario's 3-month rate and the Baseline's, in basis points, and applies it as an instantaneous shift: the quarterly NII effect is (static change + migration) / 4 at that gap. In the severe case rates fall well below Baseline, so the effect reduces NII. This is a simplification: the real curve moves gradually, and the model applies a 12-month shift every quarter.

8. Basel standardised EVE

The worst of the six Basel standardised scenarios is a parallel fall in rates, costing $32.0B, or 14.1% of Tier 1 ($227.4B), just under the 15% supervisory outlier threshold ($34.1B). Bank of America does not disclose EVE, and the result is not robust: it is 15.7% before hedges and ranges from about 9% to 23% of Tier 1 across the assumptions below, so the data cannot show whether the bank is an outlier.

Why it was built. Bank of America does not disclose EVE, so it is built from assumptions and Basel guidelines, since there is no disclosed figure to fit. EVE complements NII: NII measures the 12-month income effect, EVE the effect on the present value of banking-book cash flows.

What comes from the Basel text (BCBS d368, April 2016, Section IV and Annex 2).

- Six scenarios: parallel up, parallel down, steepener, flattener, short-rate up, short-rate down.
- USD shock sizes: parallel 200 basis points, short 300, long 150.
- Shock shape by time t in years: short shock S(t) = exp(-t/4); long shock = 1 - S(t).
- 19 time buckets with fixed midpoints (0.0028 years to 25 years).
- Non-maturity deposit (NMD) core-proportion caps of 90% (retail transactional), 70% (retail non-transactional) and 50% (wholesale); average-maturity caps of 5, 4.5 and 4 years; non-core NMDs treated as overnight.
- Prepayment multipliers of 0.8 when rates rise and 1.2 when rates fall.
- Banking book only (trading excluded); equity excluded; continuous discounting.
- The measure is the worst loss of the six scenarios, tested against 15% of Tier 1.

$$\Delta_{par} = \pm 200, \quad \Delta_{short}(t) = \pm 300\, e^{-t/4}, \quad \Delta_{long}(t) = 150\,(1 - e^{-t/4})$$

$$\Delta_{steep}(t) = -0.65\,|\Delta_{short}(t)| + 0.9\,|\Delta_{long}(t)|, \qquad \Delta_{flat}(t) = +0.8\,|\Delta_{short}(t)| - 0.6\,|\Delta_{long}(t)|$$

Valuation. For each position i the model builds a monthly cash-flow vector, assigns each cash flow to its Basel time bucket, and discounts at the bucket midpoint with a position-specific spread:

$$PV_i(\Delta) = \sum_{b=1}^{19} CF_{i,b}\; \exp\!\left(-\left[\max\!\left(r_{0,b} + \Delta_b/10^4,\; 0\right) + z_i\right] \cdot t_b\right)$$

$$\Delta EVE = \sum_{assets} \left(PV_i(\Delta) - PV_i(0)\right) - \sum_{liabilities} \left(PV_i(\Delta) - PV_i(0)\right) - \text{hedge offsets}$$

The base curve r0 uses the Fed series for 2025Q4 (3-month 3.70%, 5-year 3.70%, 10-year 4.10%), linearly interpolated between nodes. The spread z_i is solved (Brent root-finding) so that the position's base PV equals its target value: book value for most positions, Y-9C fair value for securities (this carries the roughly $80B unrealised held-to-maturity loss), and the 10-K fair value for fixed loans (about $12.7B below carrying value, allocated to fixed loans in proportion). Deposits and long-term debt stay at book. Rates are floored at zero after the shock, an assumption that Basel allows as long as the floor is not above zero.

Positions and cash-flow rules.

Position	Balance source	Cash-flow rule
Cash and balances, reverse repo	Y-9C	Floating, resetting overnight and monthly
Floating loans (66.3% of loans)	Y-9C	Floating, one-month reset
Fixed residential mortgages	Y-9C first and junior liens	Level-pay pool: 4.0% coupon, 25-year term, 6% CPR, prepayment multiplied by 0.8 or 1.2
Other fixed loans	Y-9C residual	Level-pay over 5 years, 6.5% coupon, 10% CPR
Securities up to 1 year, 1-5 years	Y-9C buckets	Uniform amortisation across the bucket at the portfolio yield
Securities over 5 years	Y-9C bucket	MBS-like level-pay, 25-year term, 7% CPR
Non-maturity deposits	Y-9C NIB, interest-bearing, foreign	Split retail and wholesale (60% retail assumed); core share and maturity at the Basel caps, spread evenly to twice the average maturity; non-core overnight
Repo, other borrowings up to 1 year	Y-9C	Floating (monthly) and uniform within the year
Long-term debt	Y-9C over-1-year borrowings	Effectively floating share 78.4%; the rest fixed, spread over the 10-K maturity ladder ($29.8B, $32.5B, $34.4B, $28.4B, $10.0B and $128.4B thereafter) at 4.0%
Subordinated debt	Y-9C	Uniform from 5 to 15 years at 4.8%

Hedges. The Y-9C does not show swaps, so the model uses the FY2025 10-K fair-value-hedge table: $175.7B of hedged long-term debt and $236.3B of hedged available-for-sale securities, where open hedges convert the hedged items to variable rate. The effective floating share of long-term debt is (31.4 contractual floating + 175.7 hedged) / (317.8 - 53.8 structured liabilities) = 78.4%. On the asset side, the hedged AFS amount is 25.9% of all

securities (61% of AFS), and its value change is subtracted from the securities' change: pay-fixed swaps lose value when rates fall, offsetting the gain on the securities. Cash-flow hedges and ALM swaps outside fair-value hedges are not included.

Results ($B change in EVE; negative is a loss).

Scenario	After documented hedges	% of Tier 1	Gross, no swaps	% of Tier 1
Parallel up	+22.8	+10.0%	+22.6	+10.0%
Parallel down	-32.0	-14.1%	-35.8	-15.7%
Steepener	-7.6	-3.3%	-5.1	-2.2%
Flattener	+9.6	+4.2%	+5.8	+2.5%
Short rate up	+14.2	+6.2%	+10.1	+4.5%
Short rate down	-4.3	-6.5%	-10.6	-4.7%

What drives the parallel-down loss. Falling rates raise the value of assets but raise the value of the deposits that are treated as long-dated even more. The main contributions are:

Position	Change in EVE ($B)
Securities over 5 years	+62.8
Fixed residential mortgages	+33.7
Securities 1-5 years	+16.9
Other fixed loans	+7.2
Retail non-transactional deposits (core)	-51.0
Retail transactional deposits (core)	-29.8
Wholesale interest-bearing and foreign deposits (core)	-27.2
Wholesale non-interest-bearing deposits (core)	-8.7
Fixed long-term debt, 5-20 years	-6.9
Subordinated debt	-5.8

The measure therefore depends mainly on how long the Basel rules let the bank treat its deposits, and on how long its assets run, which is where the assumptions bite.

Sensitivity of the worst-scenario result.

Case	Worst scenario	Worst loss ($B)	% of Tier 1
Base (after documented hedges)	Parallel down	-32.0	14.1%
Deposit core share and maturity at 75% of caps	Steepener	-27.0	11.9%
Deposit core share and maturity at 50% of caps	Parallel up	-53.1	23.4%
Retail share of domestic deposits 45%	Parallel down	-23.2	10.2%
Retail share of domestic deposits 75%	Parallel down	-40.8	17.9%
Mortgage CPR 3%	Parallel down	-22.7	10.0%
Mortgage CPR 10%	Parallel down	-40.1	17.6%

Case	Worst scenario	Worst loss ($B)	% of Tier 1
No documented hedges (gross)	Parallel down	-35.8	15.7%
Hedged share of AFS 30% (base 61%)	Parallel down	-21.4	9.4%
Hedged share of AFS 90%	Parallel down	-42.0	18.5%
Long-term debt effective floating 65% (base 78%)	Parallel down	-37.0	16.3%
Long-term debt effective floating 95%	Parallel down	-25.8	11.4%

The results straddle 15% of Tier 1 and the worst scenario changes with the deposit assumptions (parallel down in most cases, parallel up when deposits are treated as shorter-lived). The base case sits just below the line, so an interviewer or reviewer should hear the range, not only the point estimate. EVE is illustrative: the deposit split, core share, prepayment speeds and coupons are assumptions, not Bank of America data.

9. Allowance and CET1 waterfall

The waterfall turns losses and PPNR into a CET1 ratio each quarter from the 2025Q4 starting point: 11.36% in the severe case falls to 10.26% at its minimum (2027Q4) and ends at 10.30% in 2028Q1. The Fed's own figures for Bank of America are a minimum of 9.9% and 10.8% at the end. The 10.0% line (4.5% minimum + 3.0% G-SIB surcharge + 2.5% stress capital buffer floor) is used here as a reference, not as the pass/fail test: the buffer is set from the size of the decline, and the Fed's own minimum is 0.1 points below the line. The allowance and dividend steps follow the Fed's published rules; an earlier version of this document used a calibrated reserve horizon and continuing dividends, and those are kept as sensitivities in section 10.

Starting values (Y-9C 2025Q4). CET1 = reported ratio x risk-weighted assets = 11.3602% x $1,772.941B = $201.4B; loans held for investment $1,243.5B; allowance $13.2B (1.06% of loans); common dividends $8.083B a year ($2.02B a quarter) and preferred $1.445B a year; effective tax rate 19.1% (income taxes $7.186B over pre-tax income $37.695B).

Allowance (the Fed's published rule). The Fed's supervisory stress test methodology (Overview of Modeling Framework) states that the allowance at the end of each quarter covers the projected loan losses of the following four quarters, that the Fed projects 13

quarters of losses, and that any difference between its supervisory allowance at the start and the firm's actual allowance is smoothed linearly into provisions over the nine quarters. With r the scenario's annualised loss rate, L loans and t the quarter:

$$A^{sup}_t = L \sum_{h=1}^{4} \frac{r_{t+h}}{4}, \qquad A_t = A^{sup}_t - \left(A^{sup}_0 - A^{actual}_0\right)\left(1 - \frac{t}{9}\right)$$

The scenario's 13 quarters are exactly enough to give the four-quarters-ahead reserve at quarter 9, so no extension of the path is needed. On the severe path the supervisory start allowance is $13.9B against Bank of America's actual $13.2B, a difference of $0.7B that is spread over the nine quarters; that the two are close is a check on the rule. The Baseline and constructed Adverse cases apply the same rule to their own paths. Provision each quarter equals net charge-offs plus the change in the book allowance:

$$NCO_t = \frac{r_t}{4}\, L, \qquad Prov_t = NCO_t + (A_t - A_{t-1})$$

Waterfall each quarter.

$$PTI_t = PPNR_t - Prov_t - Sec_t - Trading_t - Other_t, \qquad NI_t = PTI_t\,(1 - 0.191)$$

$$CET1_t = CET1_{t-1} + NI_t - Div^{pref} - \mathbb{1}[t=1]\,Div^{common}, \qquad \text{ratio}_t = CET1_t \,/\, 1{,}772.9$$

Risk-weighted assets are held flat; the Fed's own projection is also flat ($1,772.9B to $1,771.1B). A pre-tax loss earns a tax benefit at the same rate, so no deferred-tax limit is modelled.

Capital actions (the Fed's assumption). The codified rule on capital actions in the supervisory stress test (12 CFR 252.44(c)), as I could read it through a web reader, assumes no common dividends, no repurchases and no issuances, and continued payments on preferred and subordinated instruments. I could not confirm the treatment of the first quarter from that text, so it is inferred from the numbers: the Fed's published CET1 path implies about $5.5B of dividends and other effects over nine quarters, and nine quarters of preferred dividends ($3.25B) plus one quarter of common ($2.02B) is $5.3B. The model pays preferred dividends every quarter and common only in 2026Q1, $5.3B in total. The earlier version assumed common dividends continued ($21.4B), which was the largest single reason the model's CET1 ended below the Fed's.

Fed-set stress items. Three items come from the Fed's global market shock and counterparty-default components, which are specified shocks, not forecasts, so they are taken from the Fed's published Bank of America result and spread evenly over the first nine quarters: credit losses on securities $0.7B, trading and counterparty losses $6.6B, other losses $1.9B (total $9.2B). They are applied at 100% in the severe case, 50% in the constructed Adverse case (same weight as the scenario construction), and 0% in Baseline.

Not modelled. Other comprehensive income (the Fed has +$4.7B, about +0.26 points of CET1), balance-sheet growth (the Fed also holds balance sheets constant), buybacks and issuance (none, as in the Fed's assumptions) and deferred-tax-asset limits.

Nine-quarter results (2026Q1-2028Q1).

Measure ($B unless stated)	Baseline	Constructed Adverse	Severely Adverse
PPNR	119.8	90.0	67.6
Net charge-offs	16.4	27.8	59.3
Provisions	10.4	27.5	75.3
Allowance at end	7.2	12.9	29.2
Pre-tax income	109.3	57.9	-16.8
CET1 minimum (%)	11.80 (2026Q1)	11.57 (2026Q1)	10.26 (2027Q4)
CET1 at 2028Q1 (%)	16.05	13.71	10.30

How the severe reserve behaves. The supervisory allowance at the start is $13.9B, the next four quarters of severe losses, against the actual $13.2B. The reserve then rises to $18.0B in 2026Q1, $29.4B in 2026Q3 and a peak of $39.4B in 2027Q2, as the four-quarters-ahead losses rise towards the peak of about $10B a quarter, and falls to $29.2B in 2028Q1 as forward losses ease. Provisions are $6.9B in the first quarter and peak at $10.5B in 2026Q4. The Fed's published result implies a build of only $4.6B over the nine quarters and an end reserve near $17.8B; the model's build is $16.0B (section 10). In the Baseline and constructed Adverse cases the same rule releases or builds reserve against their own paths, and the Baseline CET1 ratio rises to 16.05% because, under the Fed's capital-action assumption, no common dividends are paid after the first quarter, so the Baseline CET1 is a stress-test construct and not a forecast.

10. Results and analysis

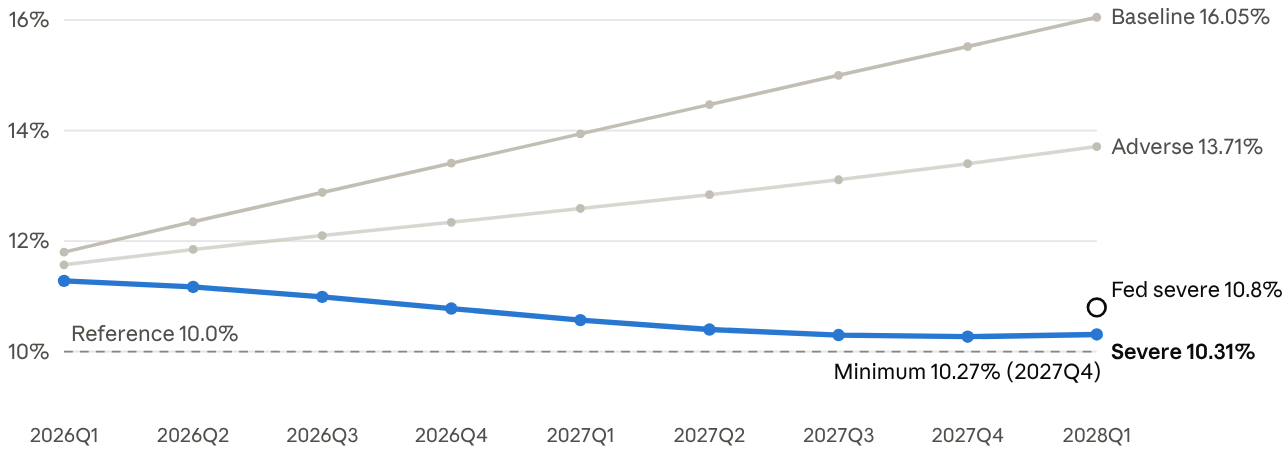
The model lands close to the Fed on PPNR ($67.6B against $66.2B) and on minimum CET1 (10.26% against 9.9%), and less closely on 2028Q1 CET1 (10.30% against 10.8%) and pre-tax income (-$16.8B against -$12.5B). Provisions are $5.8B above the Fed's. The two PPNR components that sum to a near match are individually far apart, so the total is not evidence that the income side is right.

Severe scenario against the Fed's published result for Bank of America (nine quarters, 2026Q1-2028Q1).

Measure	Model	Fed	Model minus Fed
CET1 ratio, start (%)	11.36	11.4	-0.04
CET1 ratio, minimum (%)	10.26	9.9	+0.36
CET1 ratio, 2028Q1 (%)	10.30	10.8	-0.50
PPNR ($B)	67.6	66.2	+1.4
Net interest income ($B)	127.9	151.1	-23.2
Noninterest income ($B)	115.8	89.1	+26.7
Noninterest expense, incl. operational-risk loss ($B)	176.1	174.0	+2.1
Provisions ($B)	75.3	69.5	+5.8
of which loan losses ($B)	59.3	64.9	-5.6
of which allowance build ($B)	16.0	4.6	+11.4
Securities, trading and other losses ($B)	9.2	9.2	0 (taken from the Fed)
Pre-tax income ($B)	-16.8	-12.5	-4.3

Severe CET1 bottoms at 10.27% in 2027Q4, 0.27 points above the 10.0% reference

CET1 ratio, % of risk-weighted assets, by quarter, 2026Q1-2028Q1, using the Fed's allowance and capital-action rules



cet1_waterfall.csv, 2026Q1-2028Q1; Fed end point from DFAST 2026 results (the Fed's minimum is 9.9%); Adverse is the constructed case

The constructed Adverse case stays above the reference line (minimum 11.57%) and the Baseline rises to 16.05%; the Baseline rise reflects the Fed's assumption of no common dividends after the first quarter, so it is not a forecast.

Provisions: the remaining gap is allowance build. With the Fed's published allowance rule the model's provisions are $75.3B against the Fed's $69.5B. Loan losses are $5.6B below the Fed's ($59.3B against $64.9B), so the model is not too pessimistic about losses; its allowance build is $16.0B against the Fed's $4.6B. The model's end reserve of $29.2B compares with the Fed's implied $17.8B ($13.2B opening plus $4.6B): the Fed's own loss path must be falling quickly in quarters 10 to 13, whereas the model's severe losses are still about $10B a quarter at quarter 9 and unemployment is still 9.4% in 2028Q1. The Fed's nine-quarter results file, which is the aggregate of all banks, shows losses peaking in quarter 5 and falling to about 75% of the peak by quarter 9, which is consistent with that (see the loss-timing paragraph below). The effect of each method is:

Allowance and dividend method (severe)	Provisions ($B)	End allowance ($B)	Dividends ($B)	Pre-tax ($B)	CET1 min / end (%)
Fed rules (base)	75.3	29.2	5.3	-16.8	10.26 / 10.30
Fed rules with loss timing taken from the Fed's aggregate path	63.0	16.9	5.3	-4.5	10.74 / 10.86
Fed allowance rule, common dividends continue	75.3	29.2	21.4	-16.8	9.38 / 9.38

Allowance and dividend method (severe)	Provisions ($B)	End allowance ($B)	Dividends ($B)	Pre-tax ($B)	CET1 min / end (%)
Calibrated 7.2-quarter allowance, Fed dividends	86.1	40.0	5.3	-27.7	9.40 / 9.80
Calibrated 7.2-quarter allowance, common dividends continue (earlier base)	86.1	40.0	21.4	-27.7	8.83 / 8.89
Fed published or implied	69.5	17.8	5.5	-12.5	9.9 / 10.8

An earlier version of this document said the Fed publishes no reserve horizon and used a calibrated 7.2-quarter horizon. That was wrong: the Fed's methodology states the four-quarter rule, and it is now the base. Two things support it: the supervisory start allowance on the severe path ($13.9B) is close to Bank of America's actual $13.2B, whereas the calibrated horizon had to be solved on the Baseline path to reach it; and the Fed's own balance sheet is also held constant. The 7.2-quarter variant remains as a sensitivity.

Loss timing (Fed nine-quarter file). The Fed's detailed nine-quarter file is the aggregate of all banks, not Bank of America, so it gives only the shape of losses: total loan losses peak in quarter 5 and are about 75% of the peak by quarter 9 (CRE peaks in quarter 8; cards fall to 61% of peak), and provisions fall below loan losses from quarter 5, so reserves are released in the second half. The model's severe losses peak in quarter 8 and are still 99% of the peak at quarter 9. Re-timing the model's severe path to the Fed's aggregate shape, keeping Bank of America's own $59.3B nine-quarter total and extending quarters 10 to 13 at the Fed's last quarterly decline, gives an end allowance of $16.9B (Fed implied $17.8B), provisions of $63.0B (Fed $69.5B), pre-tax income of -$4.5B (Fed -$12.5B) and severe CET1 of 10.74% minimum and 10.86% at the end (Fed 9.9% and 10.8%). It is a sensitivity and not the base, because the shape is the all-bank aggregate. It shows that most of the remaining reserve gap is loss timing, and that the model, once both rules and timing follow the Fed, is no longer pessimistic.

Loss rates by portfolio (nine-quarter cumulative, severe). Only the total is scaled to Bank of America; the segments are industry curves, shown beside the Fed's.

Portfolio	Model (%)	Fed, BAC (%)	Model minus Fed (pp)
Total loans (BAC-scaled)	4.7	5.2	-0.5
Credit cards	17.4	16.2	+1.2

Portfolio	Model (%)	Fed, BAC (%)	Model minus Fed (pp)
Commercial and industrial	7.8	6.5	+1.3
Other consumer	3.4	2.0	+1.4
Mortgage (model) against first-lien (Fed)	4.0	1.3	+2.7
Commercial real estate	1.5	9.3	-7.8

The total is within 0.5 points because large errors in opposite directions cancel: the CRE curve misses by 7.8 points (the sample has no CRE price collapse near the scenario's -38.8%, and the CRE price coefficient carries the whole response) and the mortgage curve is three times the Fed's (the industry curve reflects pre-2008 underwriting). That cancellation is a limit on how far the total can be trusted.

Diagnosing the CRE miss (diagnose_cre.py). Two checks were run on the model's own history. First, a back-test: starting from 2007Q4 and feeding in the actual 2008Q1-2010Q1 macro path, the model predicts a nine-quarter CRE loss of 2.0% against 4.4% realised, so it under-predicts the episode it was fitted on by a factor of 2.2. The cause is the log-linear partial-adjustment form: from a low starting rate (0.21% today) losses ramp up too slowly. Second, replacing the four-quarter price change with a drawdown driver (price against its trailing 12-quarter peak) fits equally well (R-squared 0.90) and lifts the severe result to between 1.0% and 2.7% depending on the specification; none of the variants is clearly better, so the base specification was kept. Correcting each specification for its own back-test shortfall gives 3.2% to 4.5%.

That leaves a gap of about 5 points to the Fed's 9.3%. For context, the worst nine-quarter loss in the industry history used here is 5.8% (2008Q4-2010Q4), so the Fed's figure is about 1.6 times the worst realised episode. That is consistent with the supervisory severity exceeding history and with bank-level loan composition, but neither can be tested from published Fed documents, so it is stated as consistent with the evidence, not shown. The conclusion for use: the CRE segment number should not be quoted, and the total loss rate is more reliable than its parts.

How much the two segment misses matter in total. Setting the CRE and mortgage curves to the Fed's values, with the same Bank of America loan mix and scale factor, changes the total nine-quarter loss as follows.

Variant	Total loss (%)	Total loss ($B)	Change ($B)
Model as built	4.74	59.2	0
CRE set to the Fed's 9.3%	5.16	64.4	+5.2
Mortgage set to the Fed's 1.3%	4.17	52.0	-7.2
Both	4.58	57.2	-2.0
Fed total	5.2	64.9	

Each error is material on its own (CRE $5.2B, mortgage $7.2B, about 9% and 12% of total losses), and they largely offset: together they move the total by only $2.0B. So the total is not resting on a large aggregate error, but the segment detail cannot be used, and a reviewer who weights CRE heavily would be right to discount the model. CRE is 5.6% of Bank of America's loans, which is why its large percentage miss moves the total less than the mortgage miss does.

Bank of America's own loan-type history (build_loan_type_history.py). The Y-9C bulk file for 2007Q1-2021Q1 (Chicago Fed, Bank of America's row) gives charge-offs and recoveries by loan type on Schedule HI-B and balances on Schedule HC-C. Charge-offs are reported year-to-date, so each quarter is differenced from the previous quarter's year-to-date figure; the rate is annualised on the average of opening and closing balances. The worst nine-quarter cumulative net charge-off rate in 2007-2021 for each portfolio, and today's balance, are:

Portfolio	Worst 9-quarter loss (%)	Window ends	Balance ($B)
CRE: construction	13.25	2010Q4	10.3
CRE: multifamily	1.43	2011Q4	7.8
CRE: non-farm non-residential	2.65	2012Q1	59.1
Mortgage: closed-end first lien	3.03	2011Q3	236.7
Mortgage: closed-end junior lien	12.57	2011Q1	1.2
Mortgage: home equity	10.29	2011Q1	26.6
Credit cards	24.54	2011Q1	106.0
Commercial and industrial	5.01	2010Q3	362.4

Three findings follow.

CRE. Weighted to today's CRE mix, Bank of America's own worst nine-quarter experience is 3.9%, against the Fed's 9.3% (2.4 times). Its whole CRE book lost 4.6% over 2008Q1-2010Q1, close to the industry's 4.4% for the same window, so the industry curve is a fair stand-in for the bank in the crisis and the model's 2.0% prediction for that window is a model-dynamics miss, not an industry-versus-bank difference. The mix has changed: construction loans, the worst performers, were a third of the book in 2007 ($34B of $102B) and are 13% now. The Fed's price fall in the severe scenario (about 39%) is similar to 2007-10, so the Fed's CRE loss cannot be reproduced from the bank's own history; it reflects supervisory severity or loan-level detail not published.

CRE refit on Bank of America's own data (refit_cre_bac.py). The CRE equation was refitted on a mix-standardised series: the bank's construction, multifamily and non-farm non-residential rates combined with today's loan-mix weights (13%, 10% and 77%), so the construction-heavy 2007-10 book does not distort it. The sample is 2007Q1-2025Q4 (60 quarters, with a start rate of 0.65% from the 2025 full-year rate) with 34 usable quarters, all from one stress episode. With the same equation and drivers as the industry model:

Specification	R-squared	Persistence	Back-test 2008Q1-2010Q1 (actual 1.86%)	Severe nine-quarter loss (%)
Unemployment and price change (as industry model)	0.88	0.52	2.50	3.68
Price drawdown only	0.82	0.74	1.10	1.85
Unemployment and drawdown	0.86	0.40	1.71	2.56
Price change and drawdown	0.84	0.74	1.94	2.76

The industry model's persistence was 0.87, which is why it ramps too slowly from a low starting rate; on the bank's own data it is 0.52 and the back-test is no longer too low (2.5% against 1.9% realised, slightly too high). The severe result is 3.8% for the matching specification and 2.3% to 3.8% across the four, close to the bank's own worst experience of 3.9% and far from the Fed's 9.3%. It is insensitive to the starting rate (3.7% to 3.8% for starting rates from 0.2% to 1.0%). Most coefficients have t-statistics below 2, so the specifications are a range, not a precise estimate. Replacing the industry CRE curve with this one raises the total nine-quarter loss from 4.77% to 4.89% (+$1.5B) and lowers severe CET1 by about 0.07 points before any reserve effect. The headline results elsewhere in this document still use the industry CRE curve; the refit is reported alongside, not wired into them, so the headline figures are unchanged. The remaining gap to the Fed, about 5.6 points, is not explained by anything in Bank of America's own history.

Current CRE book from the 2025 10-K (cre_current_book_check.py). CRE is $68.7B, up from $65.7B: office $12.4B (18.1%, down 17% in a year), industrial 19.0%, multifamily 16.0%, retail 10.1%, hotel 6.7%, multi-use 3.6% and other non-residential 25.2%. Reservable criticized loans are $8.3B (12.1%), concentrated in the 2021 and 2022 vintages (24% and 22% criticized); nonperforming loans are 2.32% (3.15% a year earlier); gross charge-offs were 0.76%; modified loans are 2.88% of the book, but those 90 or more days past due rose from $203M to $558M. The 3.8% CRE loss in the refit is about $2.6B, which is what the existing criticized loans would cost at roughly 30% severity with no migration from pass-rated loans. The Fed's $7.1B needs about 5.6% loss on the $60.4B of pass-rated loans

on top of 45% on the criticized ones, so the gap to the Fed is mostly default migration from currently healthy loans, which BAC's own history does not show. The Fed's 9.3% implies an average balance of $76B, so its CRE definition is not identical to the 10-K's.

Mortgage. First-lien loans, which are 90% of the book, lost at worst 3.0% over nine quarters in 2008-11, which included the Countrywide portfolio; weighted to today's mix (first lien, home equity and junior lien) the bank's own worst is 3.8%. The model's 4.0% therefore reproduces Bank of America's own crisis experience, and the Fed's 1.3% for first-lien loans assumes today's book performs much better than the 2008-11 book. Mortgage net charge-offs have been negative (net recoveries) since 2014. The model's mortgage figure is a judgment to anchor on the crisis, not an error in the data; the Fed's is the opposite judgment.

Reserves. Before CECL, the allowance rose from $11.6B (2007Q4) to $23.1B (2008Q4), $37.2B (2009Q4) and a peak of $41.9B, or 4.37% of loans, in 2010Q4; part of that is acquired balances (Countrywide in 2008, Merrill Lynch in 2009), which the data cannot separate. Under CECL the highest coverage was 1.96-1.97% (2020Q3-Q4), when the allowance stood at $19-20B against $4.9B of net charge-offs realised from 2020Q1 to 2021Q1. The model's end reserve under the Fed's rule, $29.2B or 2.3% of loans, sits between the CECL peak and the crisis peak; the Fed's implied $17.8B (1.4%) is below the CECL peak coverage.

PPNR: a near match from offsetting components. The model's NII is $23.2B below the Fed's and its noninterest income $26.7B above. NII is lower because balances are held flat while the Fed's path reflects balance-sheet growth, and because the rate stress takes $17.0B out of the model's Baseline NII ($144.9B). The fee gap is mainly a difference in stress severity, not in the starting level: the model's baseline fees run about 9% above Bank of America's 2025 trailing run-rate, but the Fed's severe fees are 28% below it. The reconciliation table below sets each component against the bank's own run-rate. Sensitivity of nine-quarter severe PPNR:

Assumption changed	Severe PPNR ($B)
Base overlays	67.6
NII rate effect x0.5 / x1.5	72.9 / 62.4
Fee stress x0.5 / x1.5	73.3 / 62.0
Investment banking haircut 24% / 54% (base 40%)	69.0 / 66.5
Equity pass-through 0.5 / 0.7 (base 0.6)	69.0 / 66.3

Assumption changed	Severe PPNR ($B)
Sales and trading haircut 15% (base 0%)	63.7

The range of $62B to $73B brackets the Fed's $66.2B, so the closeness of the total should be treated as partly luck.

Reconciling the PPNR components (reconcile_to_fed.py). Run-rate is Bank of America's 2025 actual quarters multiplied by 9/4, so each gap can be read as a move from today's level.

Nine quarters, $B	Run-rate	Model	Model vs run-rate	Fed	Fed vs run-rate	Model minus Fed
Net interest income	135.2	127.9	-5.4%	151.1	+11.7%	-23.2
Noninterest income	123.4	115.8	-6.2%	89.1	-27.8%	+26.7
Net revenue	258.7	243.7	-5.8%	240.2	-7.1%	+3.5
Noninterest expense	156.9	176.1	+12.2%	174.0	+10.9%	+2.1
PPNR	101.8	67.6	-33.5%	66.2	-35.0%	+1.4

Three things follow. First, net revenue agrees to within $3.5B (1.5%) and PPNR within $1.4B, so the totals that drive capital are close; the component split is where the models differ. Second, the model lowers NII and fees by similar amounts (5% and 6%), while the Fed raises NII by 12% and cuts fees by 28%. The Fed attributes its higher NII to loan growth in 2025 and to higher projected rate levels than in earlier tests; the model holds balances flat and uses Bank of America's own disclosed asset sensitivity, under which a fall in rates lowers NII, so it cannot reproduce a Fed result in which NII rises while short rates fall by 300 basis points. A possible contributor, which I have not tested, is classification: market-making activity earns net interest income that can be reported in either line, and Bank of America discloses its trading-related NII in the 10-K. Third, the $4.3B pre-tax gap is not a PPNR gap: it is $5.8B of extra provisions less the $1.4B by which the model's PPNR is higher. The component split should therefore not be quoted as validated, and the capital conclusion rests on revenue, expense and provisions in total.

CET1: where the end gap comes from. The Fed's end CET1 of 10.8% is about $191.3B against the model's $182.5B, a gap of about $8.8B. About $3.5B is the after-tax pre-tax shortfall ($4.3B at 19.1% tax) and $4.7B is the Fed's other comprehensive income, which the model leaves out; the remaining $0.6B is within the rounding of the Fed's ratios. An

earlier version of this document found a further $17B gap that it attributed to unexplained capital actions; it came from assuming common dividends continued, whereas the Fed's capital-action rule assumes they do not.

Remaining steps to the Fed's CET1. After applying the Fed's allowance and capital-action rules, the steps below show what is left, applied cumulatively to the severe scenario.

Step	Provisions ($B)	Pre-tax ($B)	CET1 min (%)	CET1 end (%)
Model with the Fed's allowance and capital-action rules	75.3	-16.8	10.26	10.30
Plus Fed other comprehensive income, +$4.7B spread evenly	75.3	-16.8	10.49	10.57
Plus loan losses set to the Fed's $64.9B	80.9	-22.4	10.29	10.31
Fed published	69.5	-12.5	9.9	10.8

Adding the Fed's OCI lifts the end ratio to 10.57%, 0.2 points under the Fed's. Raising loan losses to the Fed's level makes provisions larger, because the allowance rule scales with losses, and the end ratio returns to 10.31%. The remaining difference is therefore mainly the Fed's lower end-of-horizon reserve and the shape of its quarterly losses; the loss-timing sensitivity above removes most of it. Evenly spread OCI is a simplification, so the minimum is indicative.

What the results support. The loss side responds sensibly to the scenario (cards and C&I losses within about 1.3 points of the Fed's), the NII model is tied to disclosed figures, and the PPNR result is bracketed by sourced overlays. They do not support a capital conclusion: the segment misses cancel, the end-of-horizon reserve is $11B above the Fed's implied figure, EVE is a range straddling the supervisory line, and the PPNR match is partly offset errors.

11. Assumptions register

Every input that is assumed rather than taken from a filing, the Fed's documents or an estimate is listed here with its value, where it is used and what it affects. Items marked "tested" have a sensitivity run; the rest are open.

Assumption	Value	Used in	Why this value	Tested?
Retail share of deposits	60%	EVE (deposit classification)	Bank of America does not disclose the split in the filings used; Basel treats retail and wholesale non-maturity deposits differently	Yes, part of the 9-23% EVE range
Non-maturity deposit caps	Basel standardised caps, scale 1.0	EVE	Standard supervisory treatment; no behavioural deposit data available	Yes
Prepayment rates (CPR)	6% / 10% / 7% by loan group	EVE	Round values near industry norms; the Basel prepayment multipliers 0.8 and 1.2 then apply	Yes
Coupons	4% and 6.5% on the two fixed-rate groups	EVE	Mid-range of the observed rate environment; not disclosed by bucket	Partly
Floating reset	1 month	EVE, NII	Typical for floating commercial lending	No

Assumption	Value	Used in	Why this value	Tested?
Fixed-rate loan life	5 years	EVE	Assumed average maturity	Yes
Mortgage-backed securities prepayment	10%	EVE	Round value; replaced by an option-adjusted model in a later project	No
Floating share of loans	66.3%	EVE	Derived from the Y-9C repricing buckets	Derived
Cost of low-rate deposit tier	0.15%	NII	Anchored to the observed cost at the lowest policy rates	No
Rate floor	0	NII	Policy rate cannot go negative in the scenarios	No
Balances	Flat at the 2025Q4 level	NII, RWA, CET1	Fed's total RWA also barely moves ($1,772.9B to $1,771.1B)	Partly
Risk-weighted assets	Flat	CET1	As above	Partly
Tax rate	19.1%	Pre-tax to net income	Average of recent effective rates	No

Assumption	Value	Used in	Why this value	Tested?
Fed-set items (securities, trading and counterparty, other)	$0.7B, $6.6B and $1.9B taken from the Fed	Waterfall	These need positions and models not available outside the Fed	No
Adverse scenario	Constructed: half-way between Baseline and Severe on each driver	All outputs	The Fed's 2026 documents give only two scenarios	No
Reserve horizon	7.2 quarters	Allowance	Calibrated to the average life of the loan book	Yes (4 and 2 quarters)
Operational-risk loss	0.88% of assets, $30.0B over nine quarters	PPNR	Scaled to historical operational-loss severity	Yes
Expense elasticity	0.64	PPNR	Estimated from history	Part of the PPNR range
Fee overlays	Equity pass-through 0.6, investment banking haircut 40%, sales and trading haircut 0%	PPNR	Equity pass-through from Bank of America's 2007-09 wealth-management history; investment banking haircut from the 2022 fall in fees (section 6); the sales and trading zero is an assumption	Yes

Assumption	Value	Used in	Why this value	Tested?
Dividends	Held at the current run-rate, $21.4B over nine quarters	CET1	The Fed's capital actions are not in the documents available	No
Other comprehensive income	Zero	CET1	The Fed shows +$4.7B; no rate-driven securities model here	No
Scenario driver caps	Drivers limited to the range seen in 2000Q1-2025Q4	Loss models	No extrapolation beyond observed data	Yes (uncapped run)

How to read it. The assumptions that move the headline most are the reserve horizon (up to $13B of provisions), the fee and NII stress (about $5B each way on PPNR), dividends (an estimated $16B of the CET1 gap) and the Adverse construction. The EVE inputs matter most for the interest-rate-risk conclusion, which is why that result is given as a range.

12. Preferred sources and methods, and why

Where two routes were available, the one below was used. The rule applied throughout: prefer what Bank of America or the Fed has published over what a model infers, and prefer a method whose failure is visible over one that fails silently.

Choice	Preferred	Over	Reason
Capital and balance-sheet inputs	Y-9C regulatory filings	XBRL tags from the 10-K	Y-9C item codes are fixed and comparable across quarters; XBRL tags change names and scope

Choice	Preferred	Over	Reason
Net interest income sensitivity	30 disclosed 10-K points (Tables 44 and 45) used to fit the model	A model built from assumed betas	The fit is checked against numbers the bank itself reports
Industry loss history	FRED top-100-bank charge-off and delinquency series, 2000-2025	Bank of America's own history alone	Includes the 2008-10 cycle in full; Bank of America's segment history is too short and changed with acquisitions
Loss model form	ARDL(1,1) in one-step error-correction form	Two-step Engle-Granger error correction	The data rejected the two-step form: delinquency is near a unit root, residuals had autocorrelation of 0.85-0.97 and the error-correction terms had the wrong sign
Loss drivers	Unemployment gap, four-quarter change, BBB spread, house and CRE price growth	Unemployment level	The gap captures the turn in the cycle and avoids spurious level relationships
Standard errors	Newey-West, four lags	Ordinary	Residuals are serially correlated
Dependent variable	Log of the charge-off rate	Rate in levels	Keeps projections positive and gives proportional responses
Scaling to Bank of America	One factor (k = 0.955) fitted on 2014-2019, tested on 2020-2025	Re-fitting every coefficient to Bank of America	Few data points; one factor limits overfitting. Out-of-sample R-squared 0.72 and RMSE 0.07 percentage points

Choice	Preferred	Over	Reason
Fed result	A benchmark to compare against	A target to fit to	Fitting to it would remove the only independent check
Overlay values	Sourced from Bank of America's 2007-09 wealth-management history (pass-through) and its 2022 investment banking fees (haircut), with ranges	Judgment	Each can be traced and has a sensitivity
Interest-rate risk	Basel standardised EVE (BCBS d368)	An internal economic-value model	The standard is public and reproducible; Bank of America does not disclose EVE, so it is built from assumptions and reported as a range
Reserve method	Lifetime allowance on a 7.2-quarter horizon, with 4- and 2-quarter alternatives shown	One method presented as the answer	The Fed does not publish its method, so the sensitivity is more honest than a single point
Scenario paths	The Fed's published Baseline and Severely Adverse	Own scenarios	Makes the result comparable with the Fed's

What I would not defend. The industry-curve segments for CRE and mortgage were kept because the single-factor total scaling works at the aggregate level, but they are not good segment models (section 13).

13. Limitations

Listed from most to least consequential for the conclusions.

1. **Segment losses are wrong in opposite directions.** CRE is 1.5% against the Fed's 9.3% and mortgage is 4.0% against 1.3%. The total is close only because they offset. Any use of a segment number needs recalibration first. A back-test shows about half of the CRE miss is the model's own dynamics, and refitting on Bank of America's own history gives 3.8% (range 2.3-3.8%), but the remaining gap to the Fed's 9.3% could not be explained from the bank's history (section 10).
2. **Reserve build ($16.0B against $4.6B).** Applying the Fed's published allowance rule cut the build from $26.8B to $16.0B. Re-timing the severe losses to the shape of the Fed's aggregate path gives an end reserve of $16.9B against the Fed's implied $17.8B (section 10), so the remaining gap is mostly loss timing and not the reserve method. The shape is the all-bank aggregate, not Bank of America's, so this is a sensitivity and not the base.
3. **Capital actions and other comprehensive income are not modelled to the Fed's rules.** About $17B of the end CET1 gap and $4.7B of OCI are inferred, not measured, and the residual is uncertain by $1-2B because the Fed rounds its ratios.
4. **The PPNR match is partly luck.** NII is $23.2B under the Fed's and fees $26.7B over, because the Fed raises NII 12% and cuts fees 28% against Bank of America's 2025 run-rate while the model lowers each by about 5-6%. Net revenue agrees within $3.5B (section 10), and the $4.3B pre-tax gap is more than explained by provisions ($5.8B above the Fed's), so the component split is the weak part, not the total.
5. **NII.** The out-of-sample error on 2025 is about $0.4B, against an in-sample fit of about $0.3B; these are different tests and should not be quoted as one number. The +200bp response is overstated and the -300bp case is an extrapolation beyond disclosed points. Long-term debt floating share sits at its bound (100%).
6. **EVE is not robust.** The point estimate is 14.1% of Tier 1 (15.7% before hedges) and the sensitivity range is 9.4%-23.4%, which straddles the 15% supervisory outlier line. The result does not show whether Bank of America is an outlier.
7. **The investment banking haircut rests on one stress episode**: fees fell in 2022 (24% to 54%, depending on the base) but rose in 2008 (+7%) and 2020 (+27%), so the haircut is supported by one episode in three. A zero haircut, matching 2008 and 2020, gives severe PPNR of $71.0B against $67.6B. Sales and trading is set to zero with only a 15% sensitivity.
8. **Loan bases are not reconciled** between the FRED industry data, the Y-9C and the Fed's categories, so rates are compared on slightly different denominators.
9. **Balances and RWA are flat.** The Fed's path is also close to flat in total, but the mix is not.
10. **Adverse is constructed**, not a supervisory scenario.

11. **No independent validation.** One person built and checked everything. The companion document is the developer's own model review and documented challenge, not a validation; it lists the checks run and the findings still open.
12. **Data windows.** Fitting uses 2000Q1-2025Q4; the 2020-21 COVID dummy absorbs a period in which government support suppressed losses, so the models understate how losses behave without such support.

None of these makes the platform unusable as a demonstration of method. They do mean the numeric outputs should be presented as a reproducible exercise with stated ranges and not as a capital forecast.

14. Next steps: how to refine it

Ordered by how much each would improve the conclusions, with what it needs.

#	Refinement	What it fixes	What it needs
1	Calibrate CRE and mortgage to Bank of America's own loan-type charge-offs	The 7.8-point CRE miss (limitation 1): wire the BAC-own CRE refit (done as a side calculation) into the headline pipeline, and test a state-dependent or non-linear form on more than one stress episode. For mortgage, decide explicitly between a crisis-anchored and a current-book view	Loan-type charge-offs from the 10-K and the Y-9C schedule HC-N/HI-B across roughly 60 quarterly files
2	Replace the allowance with a method that follows the Fed's published rules (CECL lifetime on the scenario path, with reversion) and test the horizon against the 2020 and 2022 reserve paths using forecast, not realised, macro	The $22.2B reserve gap	Scenario forecasts at each historical date; the Fed's allowance disclosures

#	Refinement	What it fixes	What it needs
3	Model capital actions and OCI: dividends and buybacks per the stress-test rules, and a securities and cash-flow-hedge OCI model	The roughly $17B and $4.7B unexplained CET1 items	Securities portfolio by maturity and the Fed's capital-action rules
4	Add a balance-sheet path (loan and deposit growth by scenario)	Flat-balance NII and RWA	The Fed's published balance-sheet and RWA paths
5	Test whether trading-related NII explains the Fed's split, using the 10-K disclosure, and fit quarterly fee history	The NII and fee split against the Fed (NII 12% above run-rate, fees 28% below)	Quarterly segment fee data from 10-Q filings
6	Add more stress episodes for investment banking and sales and trading (2001-02, 2011, 2015-16, 2020, 2022)	A single-episode haircut and the zero for sales and trading	Quarterly segment revenue, which is partly screen-readable from 10-Q tables
7	Tighten EVE with behavioural deposit data and a proper repricing schedule	The 9-23% range	Bank of America's deposit-beta and run-off disclosures; cash-flow hedge detail
8	Extend NII scenarios to disclosed 2018-2024 shocks and test out of sample on more years	Extrapolation at -300bp and the bound on long-term debt floating share	More 10-K sensitivity tables
9	Fit segment models with a CRE price decline in sample (include 1990-93 where available)	The CRE price coefficient	Longer history of CRE price and charge-offs
10	Sensitivity grid across dividends, tax rate and Adverse construction	Open assumptions in section 11	Compute only

#	Refinement	What it fixes	What it needs
11	Independent review by a second person	Limitation 11	Another reviewer

Projects built on this one. Counterparty credit valuation adjustment (CVA) with a Hull-White rate model, mortgage prepayment and option-adjusted spread, and a standalone NII and EVE project each take one component here and replace its assumption with a proper model.

15. File guide, run order and sources

Run from the project folder with `py script.py`. The Basel EVE script imports the NII module, so keep all scripts in one folder.

Optional diagnostics: `py diagnose_cre.py` (after step 4) runs the CRE back-test and specification test and writes `data/models/cre_diagnostic.csv`.

Optional, after steps 5, 7 and 9: `py reconcile_to_fed.py` compares PPNR components with Bank of America's run-rate and the Fed's, and shows how the CRE and mortgage misses affect total losses (`data/waterfall/reconcile_*.csv`).

Optional, needs the Y-9C history: `py fetch_bhc_history.py` (downloads the Chicago Fed's holding-company files for 2007Q1-2021Q1 and keeps Bank of America's row, writing `data/y9c/bac_y9c_history.csv`), then `py build_loan_type_history.py` (loss rates, worst nine-quarter windows and allowance history by loan type, written to `data/models/`). Quarters from 2021Q2 on come from the per-quarter FRY9C files in `data/y9c`.

Optional, after `build_loan_type_history.py`: `py refit_cre_bac.py` refits the CRE equation on Bank of America's own mix-standardised history and writes `data/models/cre_bac_*.csv`.

Step	Script	What it does	Main output
1	fetch_fred_losses.py	Downloads FRED charge-off and delinquency series	data/fred

Step	Script	What it does	Main output
2	fetch_bac_xbrl.py	Pulls Bank of America quarterly values from SEC XBRL	bac_quarterly_xbrl.csv
3	build_calibration_panel.py	Joins losses, macro and scenarios	data/panel
4	estimate_loss_models.py	Fits the ARDL loss and delinquency models and projects scenarios	data/models
5	calibrate_bac_losses.py	Fits the single Bank of America scale factor (k = 0.955)	data/models
6	estimate_ppnr_model.py	Fits NIM, fee and expense equations, applies overlays and sensitivities	data/ppnr
7	build_nii_eve_model.py	Fits the NII model to disclosed sensitivities	data/nii
8	build_basel_eve.py	Computes the six Basel EVE scenarios and the sensitivity range	data/eve
9	build_cet1_waterfall.py	Combines PPNR, losses, allowance and capital into the CET1 path; writes the allowance sensitivity and the bridge to the Fed	data/waterfall
10	dashboard/dashboard_app.py	Streamlit dashboard of all results	Interactive app

Sources.

- Bank of America Form 10-K, 2007 to 2025: segment tables, net interest income sensitivity (Tables 44 and 45), allowance and capital.
- FFIEC Form FR Y-9C for Bank of America Corporation, quarterly (item codes in section 2).

- Federal Reserve, 2026 DFAST results (24 June 2026) and the two DFAST data dictionaries: scenario paths, Bank of America's published severe result.
- FRED: net charge-off and delinquency rates for the top 100 banks by segment; macro series.
- Basel Committee, Interest rate risk in the banking book, BCBS d368 (2016): standardised EVE.
- SEC EDGAR XBRL company facts for Bank of America.